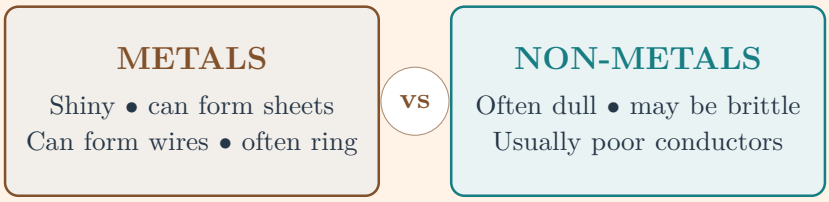
NCERT CURIOSITY • CHAPTER 4

Class 7 Science

THE WORLD OF METALS AND NON-METALS

Textbook questions first → visual notes → revision



Your learning route

Understand **malleability, ductility, sonority, conduction, rusting, oxides and real-life uses**. Then solve every NCERT end-of-chapter exercise question (pp. 55–56).

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Textbook exercise: all 12 answers

Study tip

Cover each answer first. For a "give reasons" question, write the **property** and connect it to the **observation or use**. The numbering follows NCERT *Curiosity*, Chapter 4, pp. 55–56.

Q1 • Foldable food packaging

Answer. **(i) Aluminium.** **Reason.** Aluminium can be rolled into thin sheets (malleability), is comparatively inexpensive and makes useful packaging foil.

Q2 • Metal reacting strongly with water

Answer. **(iv) Sodium.** **Reason.** Sodium is highly reactive with water and the reaction may release enough heat to ignite hydrogen. **Never test this yourself.**

Q3 • True/False with reasons

(i) **False.** Aluminium and copper are **metals**, not non-metals; they are used for utensils and statues.

(ii) **False.** The metal oxides discussed in the chapter are generally **basic**, and their solutions turn **red litmus blue**, not blue litmus red.

(iii) **True.** Oxygen is a **non-metal** and is needed for respiration in humans and many other organisms.

(iv) **False as stated.** Copper vessels are useful for boiling water because copper conducts **heat** well, not because of its electrical conductivity.

Q4 • Why only some metals make jewellery

Answer. Jewellery needs metals with suitable **lustre, malleability, ductility and resistance to corrosion**. Gold and silver suit these uses. Many other metals tarnish, corrode, react readily, or are less suitable for comfortable long-term wear.

Q5 • Match uses to jumbled names

Answer. **(i) c Copper; (ii) e Gold; (iii) a Oxygen; (iv) d Nitrogen; (v) b Chlorine.**

Use	Explanation	Substance
Wiring	Good electrical conductor	Copper
Most malleable/ductile	Can be made into foil/wire	Gold
Living things need it	Respiration	Oxygen
Fertilisers promote growth	Essential nutrient	Nitrogen
Water purification	Disinfection	Chlorine

Q6 • Burning magnesium and sulfur

Answer. Magnesium combines with oxygen to form **magnesium oxide**, a white solid; in water it gives a **basic** mixture/solution. Sulfur burns in oxygen and forms **sulfur dioxide**, which dissolves in water to produce an **acidic** solution.

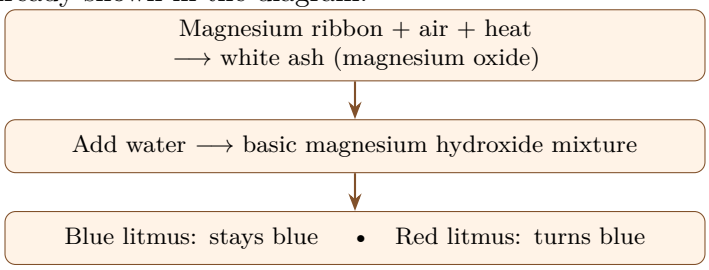
$$\text{Magnesium} + \text{Oxygen} \longrightarrow \text{Magnesium oxide}$$

$$\text{Sulfur} + \text{Oxygen} \longrightarrow \text{Sulfur dioxide}$$

Reason. The chapter contrasts typical **basic metal oxides** with **acidic non-metal oxides**.

Q7 • Complete the ash-and-litmus flowchart

Answer. Fill in: **magnesium ribbon** (starting material), **magnesium oxide + water / magnesium hydroxide mixture** (after water is added), and **blue stays blue** (blue litmus result). Red litmus **turns blue**, already shown in the diagram.



Q8 • Best pan for boiling water

Answer. **Copper** is the best choice from the given list if the main criterion is rapid heat transfer; it conducts heat very well. **Iron** can also make a durable pan. Sulfur, coal, plastic, wood and cardboard are unsuitable for the pan body because they do not combine safe high-temperature use with good conduction. A pan needs a heat-resistant handle or holder.

Q9 • Nail in oil, water or vinegar

Answer. The nail **fully protected by oil** will resist rusting because oil acts as a barrier to moisture and oxygen. Water exposure may cause rust; vinegar can speed corrosion. **Reason.** Both **water and oxygen** are needed for ordinary iron rusting.

Q10 • Properties decide uses

Answer. Examples: **copper/aluminium** conduct electricity (wires); **iron/steel** are strong (tools and structures); **aluminium** is malleable (foil); **metals** conduct heat (pans); **rubber/wood/plastic** conduct heat/electricity poorly (insulated handles); **oxygen** supports respiration; **nitrogen** is used in fertilisers; **chlorine** helps purify water.

Q11 • Could sulfur replace zinc coating?

Answer. **No.** Sulfur's low reactivity with water alone is not enough. It is **brittle** and does not form a tough, workable, lasting metal coating. A zinc layer forms a protective coating on iron (galvanisation).

Q12 • Why heat iron before shaping?

Answer. Heating makes iron **easier to deform** under hammering. The ironsmith can then shape it into tools using the metal's malleability. **It is heated for shaping, not to melt it completely.**

The ideas, explained visually

1. How to recognise materials

Property	Metals (generally)	Non-metals (generally)
Appearance	Shiny/lustrous	Often dull
Malleable	Flattened into sheets	Not malleable; often brittle
Ductile	Drawn into wires	Not ductile
Sonorous	Ring on striking	Mostly dull sound
Heat conductor	Good	Poor
Electrical conductor	Good	Poor (graphite exception)

Two words students mix up

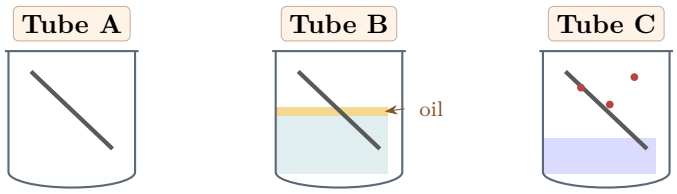
Malleable ⇒ sheets; **ductile ⇒ wires**. *S*heet and *M*alleable; *D*uctile and *D*rawn wires.

2. What happens in the experiments?

Observation → inference

1. Hammer iron/aluminium/copper vs sulfur/coal: metals can be flattened; sulfur/coal are brittle.
2. Tap a metal plate and a wooden block: ringing vs dull sounds establishes sonority.
3. Leave similar metal and wooden spoons in hot water: the metal spoon conducts heat more readily.
4. Put samples in a *low-voltage* cell–bulb tester: metals generally conduct, dry wood/plastic/rubber do not.
5. Keep clean iron nails in (A) dry air, (B) water cut off from air by an oil layer, (C) moist air: visible rust develops chiefly in **C**.

3. Rusting needs oxygen *and* moisture



Dry air — **No rust**

Water with oil barrier — **No rust**

Moist air — **Rust forms**

Protecting iron

Paint, grease, oil, or coat with zinc (**galvanisation**) to separate iron from air and water. Rusting is one type of **corrosion**.

4. Oxygen forms different oxides

Magnesium + oxygen → magnesium oxide
In water: **basic**

Sulfur + oxygen → sulfur dioxide
In water: **acidic**

Safety / common mistake

In the textbook, burning magnesium, sulfur and experiments involving sodium require a teacher. **Do not** burn substances, handle molten metal, or mix reactive metals and water on your own. The cell–bulb tester must never be connected to household mains.

Everyday uses, memory tricks and exam practice

Three memory lines

Foil → malleability; wire → ductility; bell → sonority.
Iron + air + moisture → rust.
Magnesium oxide → basic; sulfur dioxide → acidic.

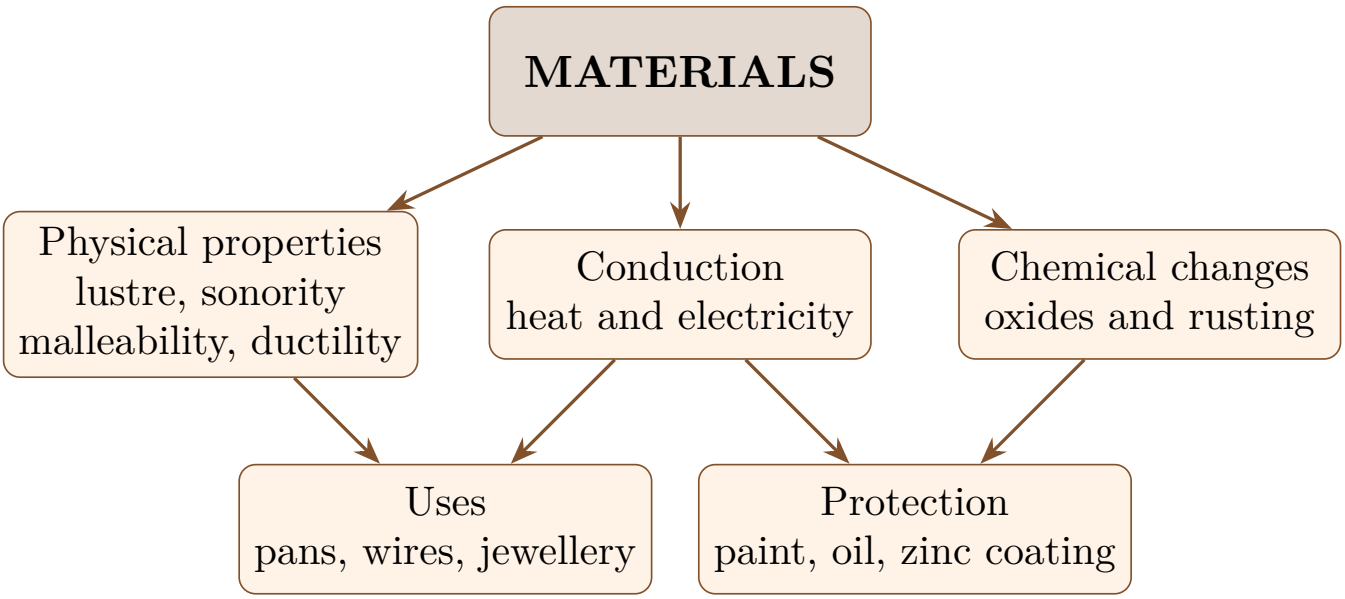
Five fast exam questions and answers

1. **What is corrosion?** Slow damage to a metal surface due to its surroundings.
2. **Why wooden handles?** Wood conducts heat poorly.
3. **Why copper wire?** Copper conducts electricity and is ductile.
4. **Why does a bell ring?** The metal is sonorous.
5. **Is every metal/non-metal identical?** No. These are general patterns, not exception-free rules.

Safety / common mistake

Do not write "copper vessels boil water because copper conducts *electricity*." The intended reason is **heat conduction**. Do not write "metal oxide turns blue litmus red": that indicates an *acid*, not the basic metal oxides demonstrated.

One-page concept map



Before the test

Can you justify each use from a property? Can you explain the three-bottle rust experiment? Can you distinguish acidic and basic oxides by their effect on litmus?